EE4-53
EE9-FPN1-02

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART IV: MEng and ACGI

Corrected copy

HVDC TECHNOLOGY AND CONTROL

Wednesday, 8 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions. Questions 1 & 2 should be answered in Booklet A and Question 3 in Booklet B

Questions 1 & 2 carry 25 marks each, Question 3 carries 30 marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	B. Chaudhuri, A. Junyent-Ferre
	Second Marker(s) :	A. Junyent-Ferre, B. Chaudhuri

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Answer ALL Questions

Question 1 and 2 carries 25 marks each, Question 3 carries 30 marks

1. a) Power reversal in an LCC HVDC link is achieved through reversal of voltage polarity. Sketch the combined rectifier-inverter characteristics of the converters at both ends and describe how the power flow through an LCC HVDC link is actually reversed in practice.

[5]

b) State the typical sequence of events following a three-phase short circuit within the AC system at the rectifier end of an LCC HVDC link. Assume that the rectifier end operates in direct current control mode while the inverter ends maintain a fixed extinction advance angle before the short circuit occurs.

[5]

c) A thermal power plant located in close electric proximity to the rectifier end of an LCC HVDC link could be prone to torsional interaction.

i) Explain the cause of possible torsional oscillations appearing on the turbine-generator shaft.

[3]

ii) State three factors that determine the severity of such torsional interaction.

[3]

d) An LCC HVDC link is planned to interconnect two 400 kV busbars A and B of an AC system where the short circuit impedances are $Z_A = 5.5\ \Omega$ and $Z_B = 6.5\ \Omega$. The reactive power provision at both ends of the HVDC link would be 55% of its active power rating under nominal voltage.

i) If the minimum required effective short circuit ratio (ESCR) is specified as 5, determine the maximum possible active power rating (in GW) of the planned LCC HVDC link.

[5]

ii) If a 6.0 GW LCC HVDC link were to be built between busbar A and B, what would be the ESCR at the two ends.

[2]

e) Explain why equidistant pulse control (EPC) is a preferred way for firing angle control for an LCC HVDC link compared to individual phase control (IPC).

[2]

2. a) Ideal no-load direct voltage (V_{d0}) is a simple way of expressing the average direct voltage in terms of the AC system voltage for LCC HVDC links.

i) State any two assumptions that are made towards quantifying the ideal no-load direct voltage.

[1]

ii) The AC-side phase-to-neutral voltages are $e_a = E_m \cos(\omega t + 60°)$ and $e_b = E_m \cos(\omega t - 60°)$ where E_m is the amplitude of the phase-to-neutral voltages. Starting from these phase-to-neutral voltages, derive an expression for the ideal no-load direct voltage of a 6-pulse line commutated converter in terms of the RMS value of the line-to-line AC-side voltage.

[4]

b) A point-to-point HVDC link interconnects two separate 400 kV AC systems. The HVDC link is rated at 2200 MW and ±600 kV and uses a standard 12-pulse converter arrangement for each pole at both ends. The resistance of each pole of the HVDC line is 3.5 Ω. The AC-side voltage at both converter stations can be assumed to be equal to the rated value. The turns ratio of the converter transformers is 0.58 with the low voltage windings on the converter side. The parameters of each 6-pulse converter used at both ends are given in Table 2.1 below:

Table 2.1: Parameters of each 6-pulse converter used at both ends of the HVDC link

Minimum firing angle	α_{min}	5°
Minimum extinction advance angle	γ_{min}	15°
Commutation resistance	R_c	6 Ω
Current margin	I_m	18%

Case I - The rectifier end of the HVDC link maintains the rated direct current while the inverter end operates with the minimum extinction advance angle. Under this situation, determine the following using the calculations on a per pole basis:

i) Direct voltage at the inverter end

[2]

ii) Firing angle (in degrees) at the rectifier

[3]

iii) Total reactive power drawn by the rectifier station

[2]

iv) Total reactive power drawn by the inverter station

[2]

v) Commutation overlap angle (in degrees) at the rectifier

[2]

Case II – If the AC-side voltage at the rectifier end reduces to 95% of its rated value, determine the following using the calculations on a per pole basis:

vi) Extinction advance angle (in degrees) at the inverter

[4]

vii) Change in total reactive power drawn by the inverter station with respect to that in Case I

[3]

viii) Commutation overlap angle (in degrees) at the inverter

[2]

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3. a) Answer the following questions related to the converter shown in Figure 3.1:

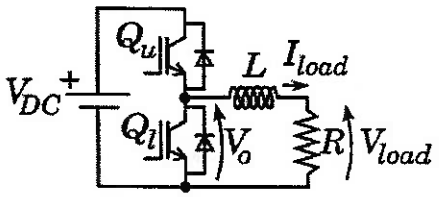


Figure 3.1 – Diagram of a DC-DC converter feeding a resistive load.

(i) Draw the waveform of the output voltage V_o over a PWM switching period. What is the relationship between the duty-cycle D of the PWM signal that drives the switches and the average of V_o (assume ideal switches)?

[2]

(ii) The transfer function between V_o and I_{load} is a first-order system of the form:

$$\frac{I_{load}(s)}{V_o(s)} = \frac{1}{Ls + R}$$

- What is the DC gain of this transfer function?

[1]

- What is the time constant of the system?

[1]

- We use a PI for a closed-loop current controller that decides what voltage, V_o, the converter must output in order to track a current reference, I_{load}*. What values should K_P and K_I have in order for the closed loop transfer function to be:

$$\frac{I_{load}(s)}{I^*_{load}(s)} = \frac{1}{\tau s + 1}$$

[3]

(iii) Consider now the case where this converter is connected to a DC grid represented by a DC voltage source as shown in Figure 3.2:

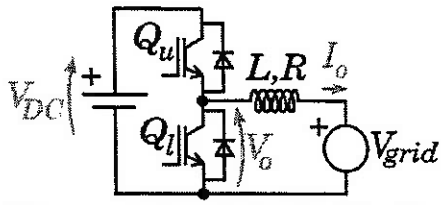


Figure 3.2 – Diagram of a DC-DC converter connected to a DC grid represented by a voltage source.

- What equation would you use to calculate the value of the current reference I_o* for a given power output reference value P*? Explain what variables you would need to measure in real time in order to be able to perform the calculation in the controller.

[2]

- Draw a block diagram showing how V_o and V_{grid} affect I_o. Show where the closed-loop current control and the feed-forward controller would appear in the diagram.

[3]

b) Answer the following questions related to the single phase inverter shown in Figure 3.3:

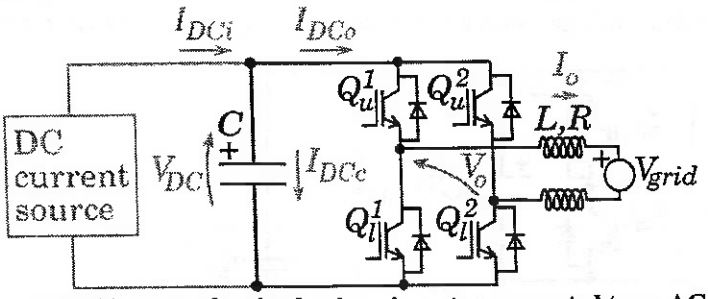


Figure 3.3 – Diagram of a single phase inverter connected to an AC grid

Parameter	Value	Units	Brief description
V_{grid}	230	V rms	Rated AC grid voltage
P_{rated}	1	kW	Rated active power delivered to the grid
f_{rated}	50	Hz	Rated frequency of the AC grid
R	0.25	Ω	Equivalent resistance of bridge plus the filter
L	15	mH	Inductance of the filter

Table 3.1 – Diagram of a single phase inverter connected

(i) Calculate the base impedance given the converter characteristics and calculate the impedance of the output filter in per-unit.

[2]

(ii) What is the capacitance of the DC link if the DC link voltage, V_{DC} is 400 V and the inertia time of the DC link, H, is 50 ms?

[2]

(iii) Consider the operating point where the converter is delivering rated power to the AC grid with unity power factor (ie $P_{grid}=P_{rated}$, $Q_{grid}=0$). Answer the following questions:

- What is the output current, I_o, and the voltage, V_o the converter must apply?

[1]

- What are the conduction losses of the converter? How much reactive power does the filter consume?

[1]

(iv) Consider now the operating point where the converter is delivering 0 apparent power to the grid. What voltage, V_o, does the converter need to apply? What would the minimum voltage of the DC link V_{DC} be in order to be able to generate such voltage?

[3]

c) Consider a basic one-arm modular converter connected between an AC grid and a DC grid represented as equivalent voltage sources as shown in Figure 3.4. Answer the questions below:

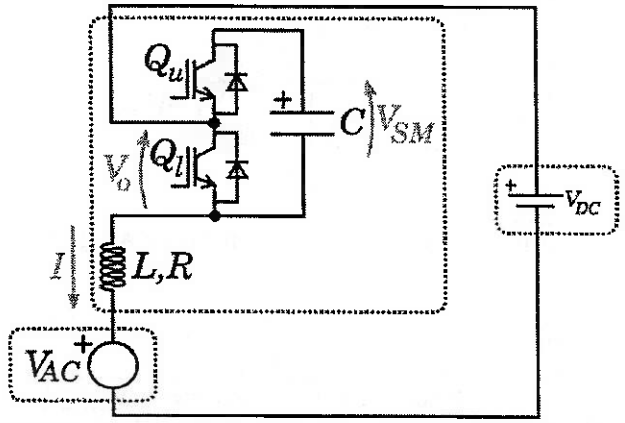


Figure 3.4 – Diagram of a one-arm modular AC/DC converter

Parameter	Value	Units	Brief description
V_{DC}	750	V	Rated DC grid voltage.
V_{AC}	398	V rms	Rated AC grid voltage
f_{rated}	50	Hz	Rated frequency of the AC gr
R	10	mΩ	Equivalent resistance of bridge plus the filter
L	0.34	mH	Inductance of the filter

Table 3.1 – Diagram of a single phase inverter connected

(i) Calculate the AC and DC components of the output current, I, and the output voltage of the arm, V_o, for a stable operating point (0 net power balance across the submodule) when the power delivered to the AC grid is P_{AC}=100 kW, Q_{AC}=0 kVAr.

[4]

(ii) Draw the approximate waveform of the output voltage, V_o, at zero power (I=0). Explain how high V_{SM} must be and what the ratio between V_{DC} and V_{AC} must be for the converter to be able to operate under this specific condition.

[3]

(iii) List the potential advantages and disadvantages of designing this converter to use three submodules in the arm compared to using a single submodule.

[2]

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